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An aromatic ring linked to the β -lactam ring has been reported to participate in hydrophobic interactions with β -lactamase active sites, and an aromatic ring bonded to the nitrogen atom of the β -lactam has shown to be beneficial to biological activity. Researchers synthesized a group of penicillanic acid analogues with substituted aromatic rings bonded to both the carbon scaffold of the β -lactam ring and to its nitrogen heteroatom to overcome the challenge of antibiotic resistance. Various substituents, including $-\text{OCH}_3$ and $-\text{NO}_2$, were utilized as groups R^1 , R^2 , and R^3 to make several analogues. An overview of the synthesis of these analogues is shown in Figure 1, where the key step is the coupling of compounds **2** and **4** driven by dicyclohexylcarbodiimide (DCC) to form a compound with two linked β -lactam rings.

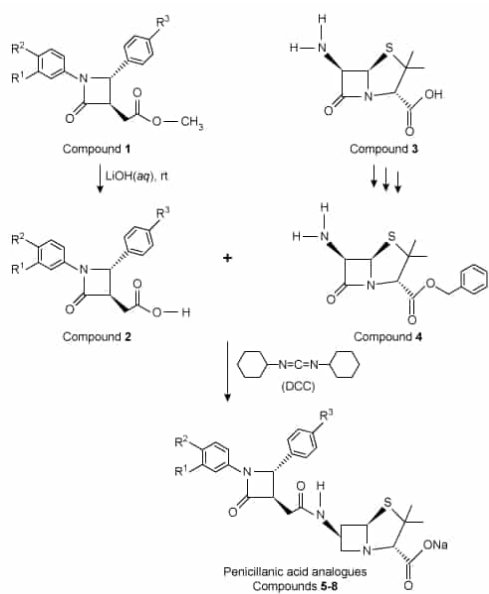


Figure 1 Synthesis of penicillanic acid analogues

The researchers studied the analogues' structure–activity relationship and antimicrobial activity against the gram-negative bacteria *Escherichia coli* and the gram-positive bacteria *Staphylococcus aureus*. Ampicillin, a penicillin derivative used to treat illnesses brought about by gram-positive and gram-negative bacteria, was used as a control. The minimal inhibitory concentrations necessary for the compounds to inhibit growth and kill 100% of the bacteria (MIC_{100}) are shown in Table 1.

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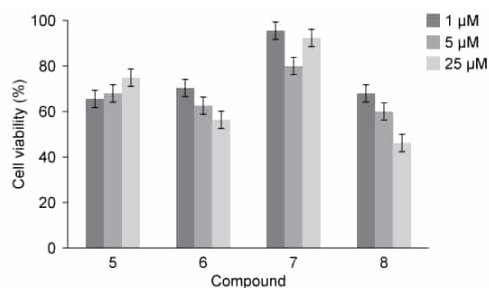


Figure 2 Cell viability assay results

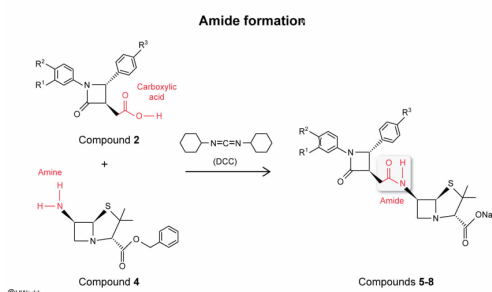
Which functional group forms during the key step of the synthesis of the penicillanic acid analogues that couples compounds **2** and **4**?

- ☐ A. Ester [4%]
☐ B. Anhydride [3%]
☒ C. Amide [81%]
☐ D. Enamine [10%]

Correct

82% answered correctly

Explanation:



The **carboxylic acid (CA) derivatives** are formed through **condensation** of a CA (or its derivative) and another molecule. The **amide** functional group, consisting of an **acyl group** bonded to an $-NR_2$ substituent (R = any alkyl group or H atom), is **formed** when a **primary or secondary amine attacks** the electrophilic **carbonyl** carbon of a CA derivative, forming a tetrahedral intermediate. The carbonyl reforms when a **leaving group** is **eliminated** from the intermediate. The negatively charged leaving group then **abstracts a proton** from the positively charged nitrogen atom. **Tertiary amines** cannot form amides because they do not contain any protons bonded to the nitrogen atom.

Amide synthesis by direct reaction with a CA is not practical because it requires extremely high temperatures. To avoid the use of high temperatures, additional reagents (ie, activators) can be used to push the reaction forward. These activator reagents convert the CA to a more reactive CA derivative, which can then react with an amine. In Figure 1, the CA group in Compound **2** **reacts with the activator** DCC to form an ester-like intermediate, which reacts with the amine in Compound **4** to form the amide linkage in compounds **5–8**.

(Choice A) An **ester** is a CA derivative in which an acyl group is bonded to an $-OR$ group rather than an $-NR_2$ group. It is made from a CA and an alcohol.

(Choice B) An **anhydride** is a CA derivative, but it contains two carbonyl groups linked together by an oxygen atom and is made by the condensation of two CAs.

(Choice D) An **enamine** is a functional group that contains nitrogen linked to a carbon that is double-bonded to another carbon. Enamines are formed by the addition of a secondary amine to an aldehyde or ketone. As such, an enamine is not a CA derivative.

Educational objective:

Amides are carboxylic acid (CA) derivatives made up of an acyl group bonded to an -NR_2 substituent.

These compounds form from the condensation of any CA derivative with a primary or secondary amine.

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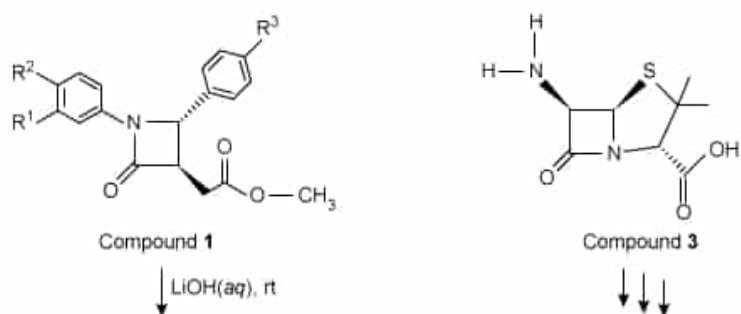
Subject:
Organic Chemistry

QID: 401269

System:
Functional Groups and Their Reactions

The β -lactam scaffold is an important feature in a class of broad-spectrum antibiotics that includes the penicillins, cephalosporins, and monobactams. These antibiotics are used to treat a variety of diseases caused by bacteria. The penicillins work by inhibiting a step in the synthesis of peptidoglycan. However, the development of antibiotic resistance in the form of enzymes such as β -lactamase is a growing problem.

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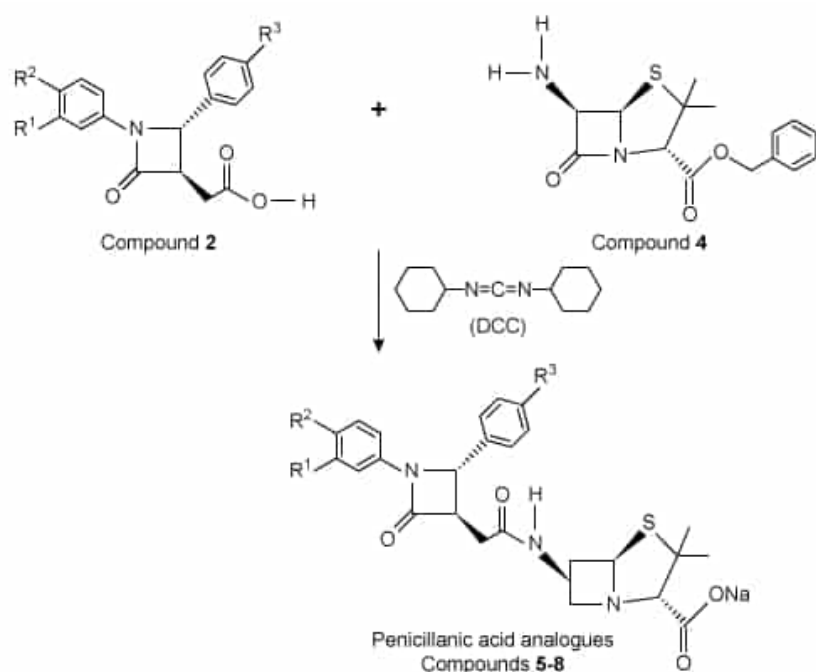


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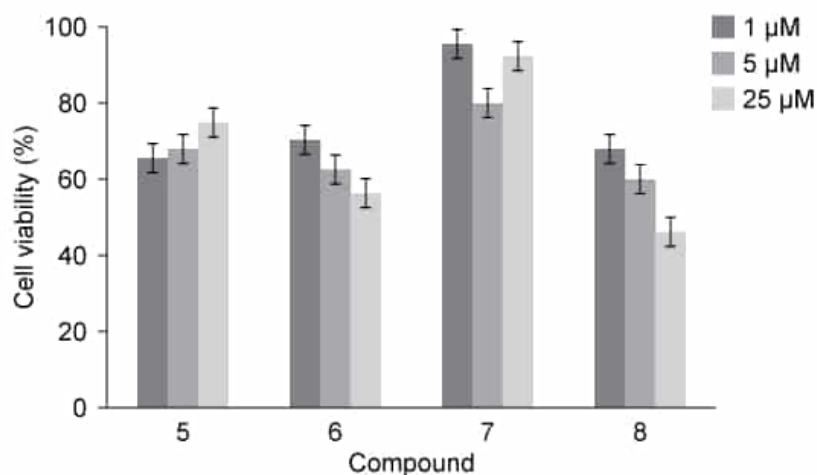


Figure 2 Cell viability assay results

In spite of the ring strain in the β -lactam ring, researchers observed a nearly quantitative conversion of Compound **1** to Compound **2** when treated with LiOH(aq) at room temperature. Which of the following reasons best explains why the ester is hydrolyzed while the amide functionality of the β -lactam does not react?

- ✔ ☒ A. The $-\text{OCH}_3$ group of the ester exerts a strong inductive effect that withdraws electron density and makes the C atom of the ester $\text{C}=\text{O}$ more electrophilic. [75%]
- ☐ B. The N atom of the amide exerts a moderate inductive effect that donates electron density to the C atom of the amide $\text{C}=\text{O}$ and makes the bond more

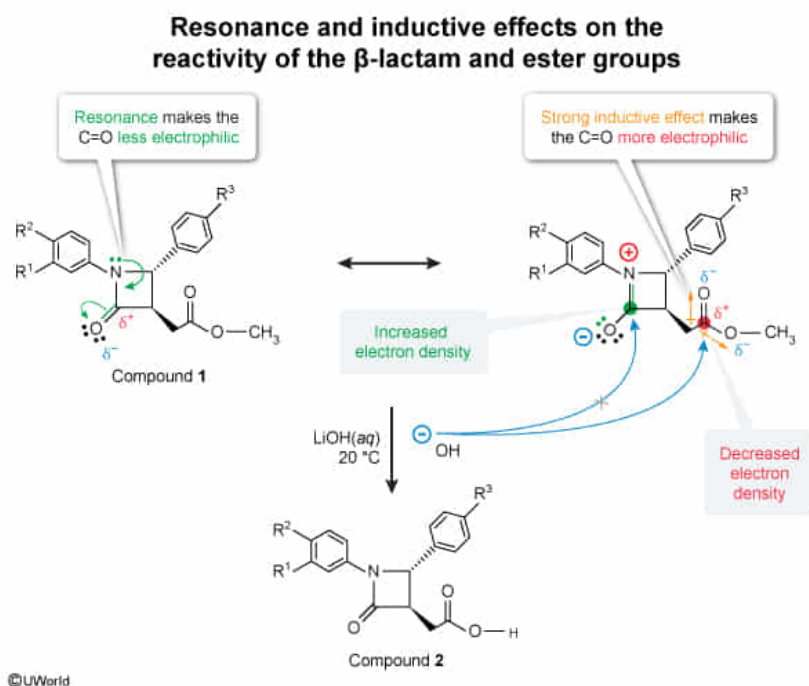
electrophilic. [6%]

- ☐ C. The $-\text{OCH}_3$ group of the ester donates electron density to the C atom of the ester $\text{C}=\text{O}$ through resonance, making the bond less electrophilic. [9%]
- ☐ D. The N atom of the amide withdraws electron density from the C atom of the amide $\text{C}=\text{O}$ through resonance, making the bond more electrophilic. [8%]

Correct

75% answered correctly

Explanation:



Carboxylic acid derivatives have varying reactivity depending on the **substituent** bonded to the **carbonyl carbon** and other factors, such as inductive effects and resonance. An **inductive effect** is an electronic property in which **electronegative** atoms or **electron-withdrawing substituents** pull electron density away from an adjacent atom toward

themselves through **sigma (σ) bonds**, creating a **dipole** with partial positive (δ^+) and partial negative (δ^-) charges at each end. **Resonance** delocalizes pi (π) bonds and **nonbonding electrons** to different parts of a molecule.

Esters and **amides** can both undergo **hydrolysis**; however, due to carbonyl reactivity differences, different conditions are required for each to be hydrolyzed. Figure 1 in the passage shows that when Compound **1**, which contains a β -lactam (cyclic amide) and an ester, reacts with LiOH(aq) at room temperature, the **ester is hydrolyzed** preferentially, forming Compound **2**. Comparing the ester $-\text{OCH}_3$ group and the amide nitrogen atom in Compound **1**:

- Ester $-\text{OCH}_3$ group
 - **Electron withdrawing** through an **inductive effect**, creating a dipole
 - **Increases electrophilicity** and reactivity of the ester's carbonyl carbon
- Amide nitrogen atom
 - **Less electron withdrawing** due to **smaller electronegativity difference** between C and N compared to C and O in esters (ie, 0.5 difference versus 1.0 difference)
 - **Resonance** with the $\text{C}=\text{O}$ bond **increases electron density** around the C atom
 - **Decreases electrophilicity** and reactivity of

carbonyl carbon

Therefore, the ester is more reactive than the amide in Compound **1** and can undergo hydrolysis under mild reaction conditions (eg, room temperature).

(Choice B) The N atom of the amide does exert a moderate inductive effect, but this effect *withdraws* electron density from the C atom of the amide C=O. Resonance counteracts this effect and makes the amide less prone to hydrolysis.

(Choices C and D) Both the -OCH_3 group on the ester and the amide nitrogen atom can be electron-*donating* through [resonance](#), but electron donation tends to make the C atom in the C=O *less* electrophilic (and less reactive); therefore, these choices do not explain why the ester reacts instead of the amide.

Educational objective:

Carboxylic acid reactivity depends on the electron-withdrawing and electron-donating capabilities of the carbonyl carbon substituents, which can act via resonance or inductive effects. Esters are more reactive than amides due in part to a stronger electron-withdrawing, inductive effect caused by the greater electronegativity difference between the C and O atoms in esters compared to the C and N atoms in amides.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401270

System:
Functional Groups and Their Reactions

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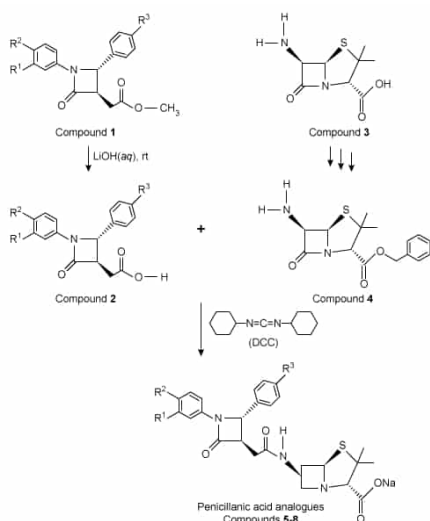


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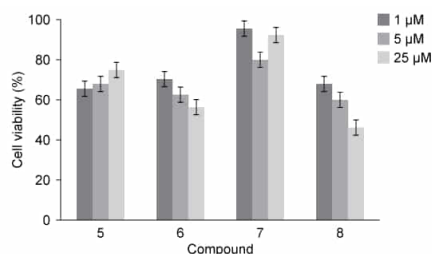


Figure 2 Cell viability assay results

Compared to noncyclic amides, β -lactams like those described in the passage are more reactive because:

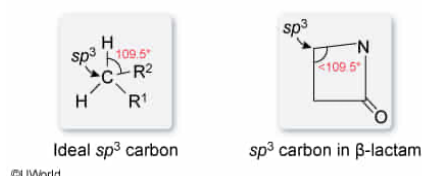
- ☐ A. the nitrogen atom in a β -lactam is sp^2 hybridized. [7%]
- ✓ ☒ B. the angles in a β -lactam ring are less than 109.5° . [59%]
- ☐ C. a β -lactam nitrogen atom has trigonal planar geometry. [5%]
- ☐ D. β -lactams can participate in more resonance. [26%]

Correct

60% answered correctly

Explanation:

Ring strain in β -lactams



β -Lactams are a class of four-membered [cyclic amides](#), and various factors contribute to their enhanced reactivities relative to noncyclic amides.

• Structural features that make noncyclic amides less reactive:

The nitrogen atom in a noncyclic amide has an sp^2 hybridized character because it donates electron density to the carbonyl carbon through [resonance](#). This resonance gives the amide C–N bond a partial double-bond character and greater stability by better distributing electron density and delocalizing charges. The partial double-bond character of the C–N bond makes the O=C–N bonds coplanar. In addition, noncyclic amides are less reactive because the --NH_2 group is a base, and therefore not a [good leaving group](#).

• Structural features that make β -lactams more reactive:

The small ring size of the β -lactams found in the penicillanic acid analogues described in the passage forces the bond angles to be smaller than the 109.5° normally observed in [\$sp^3\$ hybridized atoms](#), causing significant **ring strain** that prevents the nitrogen atom in the β -lactams from being coplanar with the C=O bond. This causes the nitrogen atom to take on a [trigonal pyramidal geometry](#), which decreases the resonance because the orbitals do not overlap very well. The resulting decrease in double-bond character gives the nitrogen atom in a β -lactam a greater sp^3 hybridized character, and the decreased donation of electron density to the carbonyl carbon via resonance makes the C=O bond more reactive.

Therefore, the **bond angles smaller than 109.5°** (ie, ring strain) found in β -lactams cause the carbonyl carbon to be **more reactive** than it would be in a noncyclic amide.

(Choices A, C, and D) β -lactam's four-membered ring structure causes the nitrogen atom to have a trigonal *pyramidal* geometry, reducing resonance. This reduced resonance gives the nitrogen an sp^3 hybridization because the nitrogen does not have partial double-bond character.

Educational objective:

β -Lactams are a class of cyclic amides that exhibit significant ring strain due to their small ring size, making the carbonyl carbon more reactive than noncyclic amides. The nitrogen atom in a β -lactam is trigonal pyramidal rather than coplanar with the carbonyl and has decreased resonance. Unlike noncyclic amides, the nitrogen atom is sp^3 hybridized.

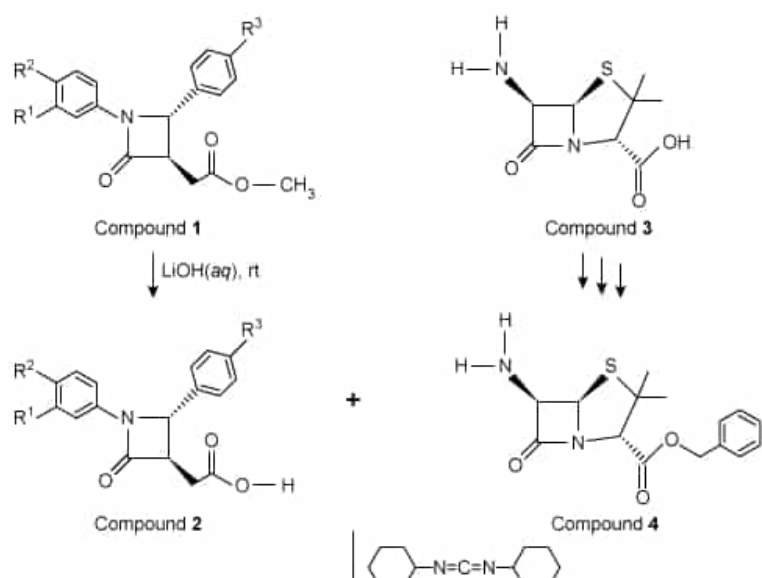
Time spent: 0 sec
Subject:
Organic Chemistry

QID: 401271
System:
Functional Groups and Their Reactions

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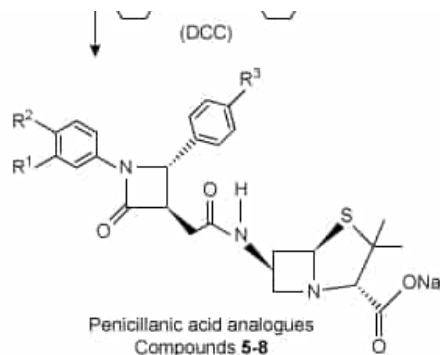


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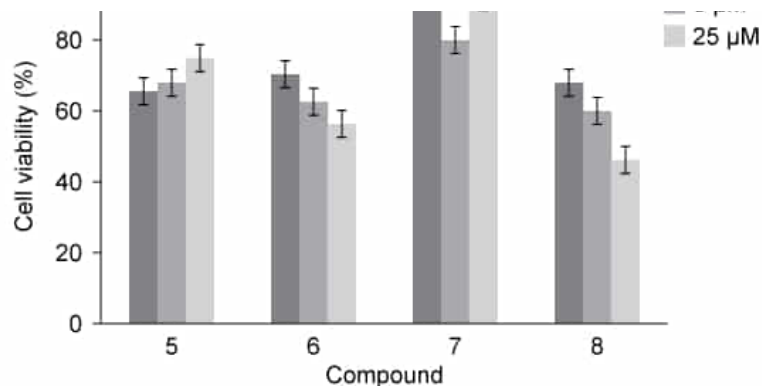
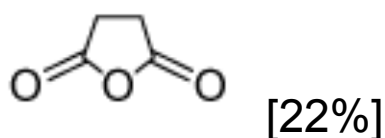


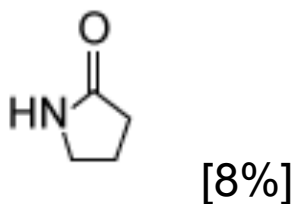
Figure 2 Cell viability assay results

The synthesis of Compound **1** begins with the nucleophilic acyl substitution of a carboxylic acid derivative with methanol under reflux conditions. This reaction forms a derivative of butanoic acid where carbon 4 is a methyl ester. Which structure was most likely the carboxylic acid derivative used as the reactant in the nucleophilic acyl substitution?

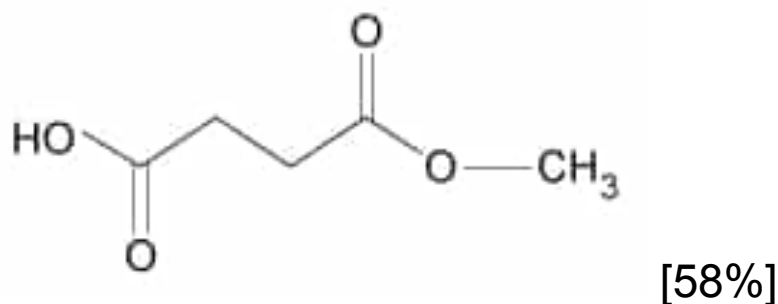
✓ ☒ A.



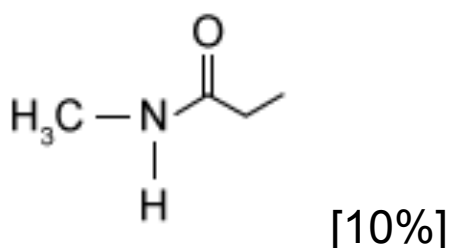
☐ B.



☐ C.



☐ D.



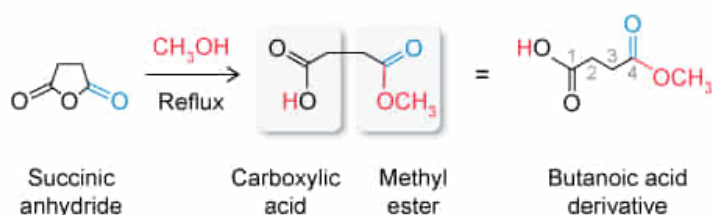
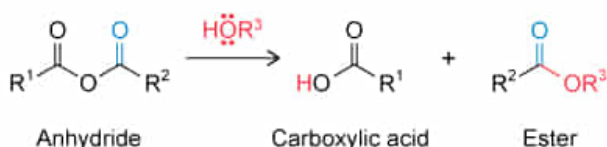
Correct

21% answered correctly

Explanation:

Nucleophilic acyl substitution

- Anhydride cleavage to ester and carboxylic acid
- Nucleophilic attack of carbonyl by alcohol
- Carboxylic acid is leaving group



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Carboxylic acid (CA) derivatives can undergo **nucleophilic acyl substitution** under acidic or basic conditions. A **nucleophile**, such as an alcohol or an amine, attacks the carbonyl carbon of the CA derivative, forming a **tetrahedral intermediate**; the leaving group is then removed.

Anhydrides within a carbon chain undergo nucleophilic acyl substitution and are **cleaved** into two molecules: a **carboxylic acid** and either an **ester or amide**, depending on whether the nucleophile is an alcohol or an amine, respectively. In contrast, cyclic anhydrides are not cleaved into two molecules but instead the ring is broken, placing the CA group and the ester (or amide) at opposite ends of the new molecule.

The reaction described in the question forms a CA and an ester, so the starting material must be an anhydride. The **prefix** of the product's parent name is *but-* (ie, the product is a butanoic acid

derivative), indicating a four-carbon chain or ring. Because the reaction forms an ester at carbon 4 on the same molecule as the CA (instead of forming two separate molecules), the original molecule must have been a cyclic anhydride with four carbon atoms, as shown in Choice A.

(Choices B and D) Although amides and lactams (ie, cyclic amides) are CA derivatives, and these structures do contain the correct number of carbons, amides cannot react with methanol to form a CA and an ester.

(Choice C) This molecule is the *product* (ie, a derivative of butanoic acid with a methyl ester at carbon 4), not the CA derivative used as the *reactant* in the nucleophilic acyl substitution described in the question.

Educational objective:

Carboxylic acid (CA) derivatives can undergo nucleophilic acyl substitution. Anhydrides are cleaved in this reaction into a CA and an ester or an amide. Cyclic anhydrides are converted into a single hydrocarbon chain with a CA on one end and an ester or an amide on the other.

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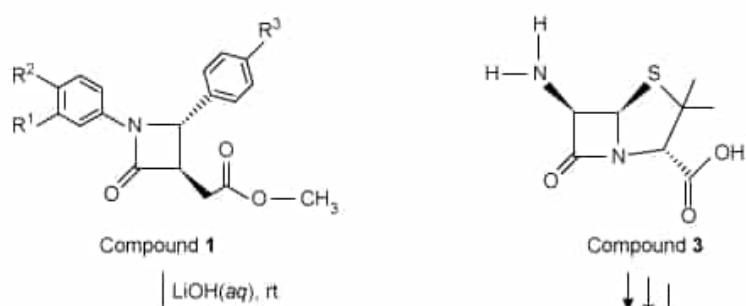
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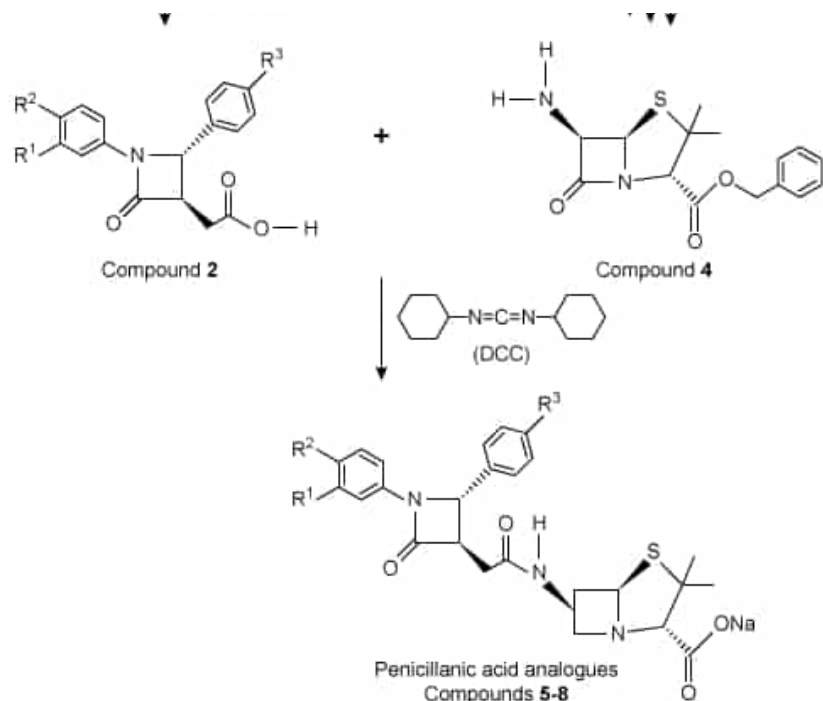


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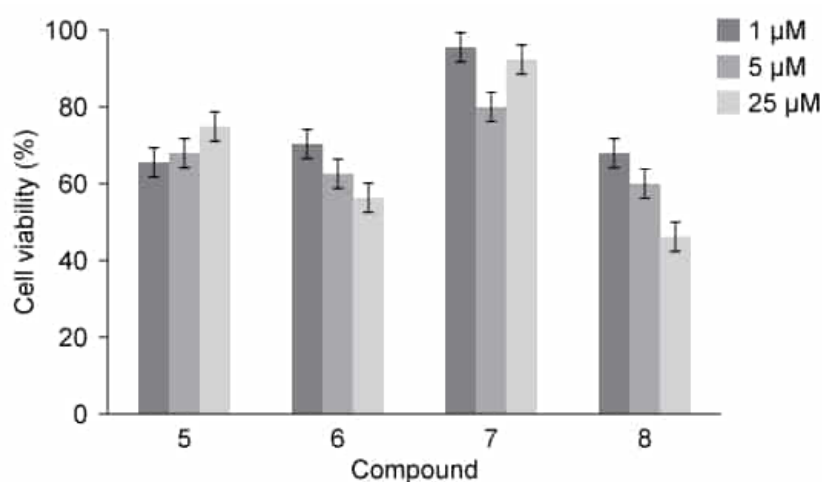


Figure 2 Cell viability assay results

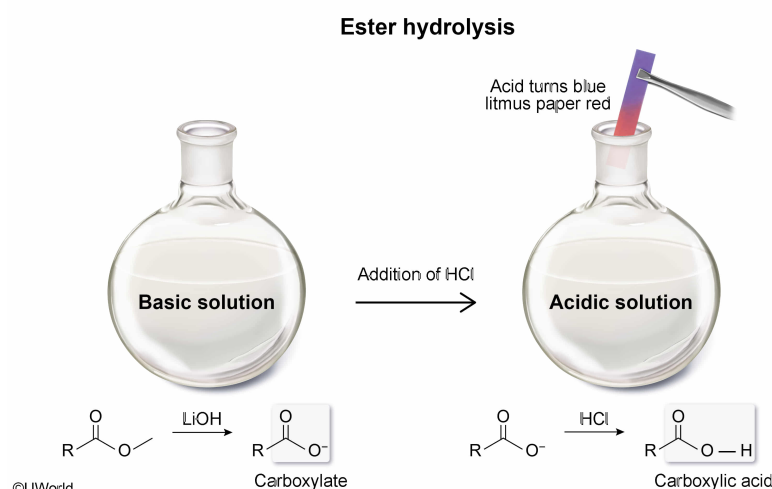
The hydrolysis of Compound **1** produces a carboxylate ion, which is quenched by the addition of HCl. What is the expected result if blue litmus paper is immersed in the fully quenched solution? The blue litmus paper will:

- ✔ ☒ A. turn red because the carboxylate is protonated. [69%]
- ☐ B. turn red because the carboxylate is deprotonated. [8%]
- ☐ C. remain blue because the solution is acidic. [7%]
- ☐ D. remain blue because the solution is basic. [13%]

Correct

69% answered correctly

Explanation:



Litmus paper is coated with litmus dye that changes color depending on the pH of the solution in which it is immersed; it is used as a **pH indicator** to determine whether a solution is acidic or basic. An **acidic** ($\text{pH} < 7$) solution changes blue **litmus paper to red** whereas a basic ($\text{pH} > 7$) solution turns red litmus paper blue.

Hydrolysis of an ester with the base LiOH creates a carboxylate ion that has a $\text{pH} > 7$ because LiOH is a strong base. **Protonation** of the **carboxylate** ion requires the **addition** of an **acid**, such as HCl . The addition of an acid to the basic solution causes the pH of the reaction mixture to decrease, becoming more acidic. Carboxylic acids are weak acids with pK_a values near 4, so the pH of a solution of carboxylate ions must be brought below 4 to fully protonate the ions.

The pH of this mixture can be monitored using litmus paper. When enough acid is added to the reaction, the carboxylate

ions are protonated, and blue litmus paper immersed in the fully quenched reaction mixture will turn red.

(Choice B) The blue litmus paper will turn red; however, the carboxylate was already deprotonated before quenching, and the addition of an acid to the reaction will *protonate* the carboxylate.

(Choice C) The solution is acidic because the addition of the strong acid HCl decreases the pH of the reaction; however, an acidic solution will turn blue litmus paper red.

(Choice D) Blue litmus paper remains blue in basic solutions. The initial reaction mixture is basic because LiOH is used for the ester hydrolysis, but it becomes acidic with the addition of HCl.

Educational objective:

Litmus paper is used as a pH indicator. In acidic solutions ($\text{pH} < 7$), blue litmus paper turns red; in basic solutions ($\text{pH} > 7$), red litmus paper turns blue. The addition of an acid to a solution protonates bases such as carboxylate ions.

Time spent:0 sec

Subject:

Organic Chemistry

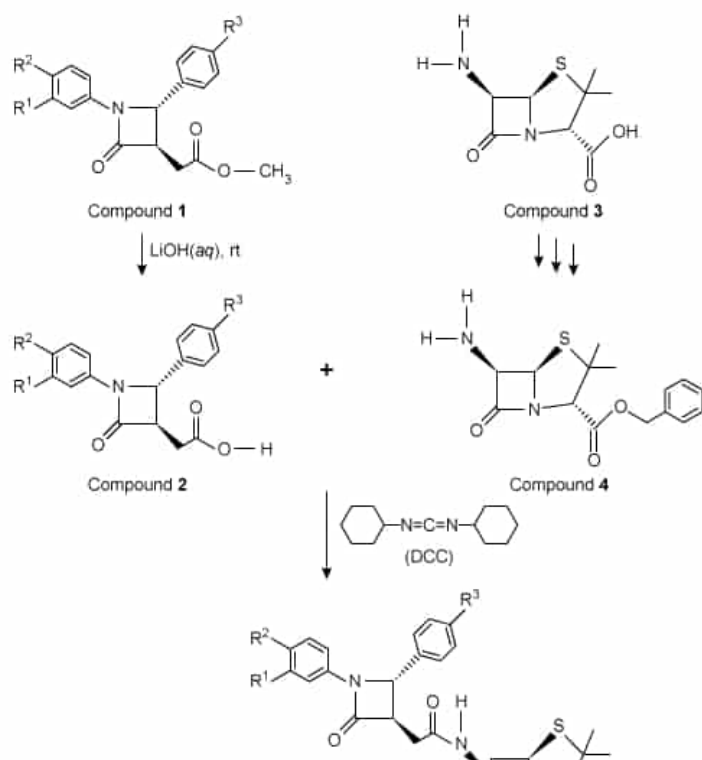
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System:

Functional Groups and Their Reactions

The β -lactam scaffold is an important feature in a class of broad-spectrum antibiotics that includes the penicillins, cephalosporins, and monobactams. These antibiotics are used to treat a variety of diseases caused by bacteria. The penicillins work by inhibiting a step in the synthesis of peptidoglycan. However, the development of antibiotic resistance in the form of enzymes such as β -lactamase is a growing problem.

An aromatic ring linked to the β -lactam ring has been reported to participate in hydrophobic interactions with β -lactamase active sites, and an aromatic ring bonded to the nitrogen atom of the β -lactam has shown to be beneficial to biological activity. Researchers synthesized a group of penicillanic acid analogues with substituted aromatic rings bonded to both the carbon scaffold of the β -lactam ring and to its nitrogen heteroatom to overcome the challenge of antibiotic resistance. Various substituents, including $-\text{OCH}_3$ and $-\text{NO}_2$, were utilized as groups R^1 , R^2 , and R^3 to make several analogues. An overview of the synthesis of these analogues is shown in Figure 1, where the key step is the coupling of compounds **2** and **4** driven by dicyclohexylcarbodiimide (DCC) to form a compound with two linked β -lactam rings.



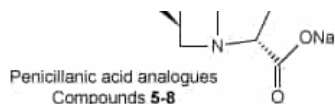


Figure 1 Synthesis of penicillanic acid analogues

The researchers studied the analogues' structure–activity relationship and antimicrobial activity against the gram-negative bacteria *Escherichia coli* and the gram-positive bacteria *Staphylococcus aureus*. Ampicillin, a penicillin derivative used to treat illnesses brought

about by gram-positive and gram-negative bacteria, was used as a control. The minimal inhibitory concentrations necessary for the compounds to inhibit growth and kill 100% of the bacteria (MIC₁₀₀) are shown in Table 1.

Table 1 Minimal Inhibitory Concentration (MIC)

Compound	MIC ₁₀₀ (μM)	
	<i>E. coli</i>	<i>S. aureus</i>
Ampicillin	69	2.9
5	>255	1.1
6	>242	2.0
7	>241	>2.0
8	>222	1.7

The penicillanic acid analogues were tested for their effect on mammalian cell viability because toxicity to cells is also an important factor to consider for potential drug candidates. The general acceptable level of cell viability for a drug candidate is 70%. These results are shown in Figure 2.

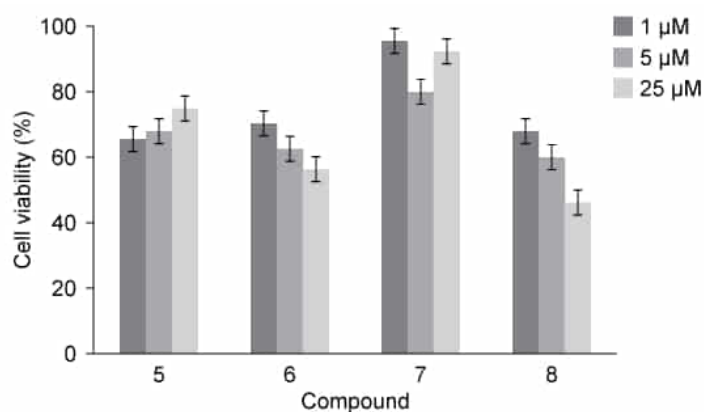


Figure 2 Cell viability assay results

Which conclusion about the bioactivity of Compounds **5–8** is best supported by the data in Table 1 and Figure 2?

- ☐ A. The addition of substituted aromatic rings to the penicillin core is detrimental to cell viability. [3%]
- ☐ B. Compound **7** has the largest effect on cell viability. [23%]
- ☒ C. Compound **5** demonstrates the greatest antimicrobial activity against *S. aureus*. [60%]
- ☐ D. Compared to ampicillin, Compounds **5–8** display an increased antimicrobial activity against gram-negative bacteria. [12%]

Correct

60% answered correctly

Explanation:

Bioactivity of compounds 5-8

Compound	MIC ₁₀₀ (μM)	
	<i>E. coli</i>	<i>S. aureus</i>
Ampicillin	69	2.9
5	>255	1.1
6	>242	2.0
7	>241	>2.0
8	>222	1.7

Compound **5**
Lowest MIC₁₀₀ concentration and
therefore has the **greatest activity**
against *S. aureus*

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Table 1 shows **minimal inhibitory concentration (MIC)** values for compounds **5–8** and the control ampicillin against *E. coli* and *S. aureus*. The MIC gives the **lowest concentration** necessary for a compound **to inhibit** the **growth** of a bacterium. The subscript "100" indicates that 100% of the bacteria are killed at that particular concentration. All the penicillanic acid analogues (compounds **5–8**) are more active than ampicillin against *S. aureus* bacteria because a lower concentration of each is needed to inhibit the growth of *S. aureus*. Of the four penicillanic acid analogues, Compound **5** has the **lowest MIC₁₀₀** value of 1.1 μM, demonstrating the **greatest antimicrobial**

activity of the tested compounds against *S. aureus*.

(Choice A) Figure 2 shows that compounds **5–8** had varying effects on cell viabilities, ranging from 45%–95%. The passage states that 70% cell viability is generally acceptable for drug candidates. Because cell viability was above 70% for some of the compounds at the tested concentrations, it can be concluded that the addition of the substituted aromatic rings on compounds **5–8** is *not always* detrimental to cell viability.

(Choice B) Compound **7** has the *smallest* effect on cell viability because, according to Figure 2, the highest percentage of cells remained alive at the three tested concentrations of Compound **7**.

(Choice D) Table 1 shows that compared to ampicillin, compounds **5–8** display *decreased* antimicrobial activity against gram-negative bacteria (ie, compounds **5–8** required much larger concentrations than ampicillin to inhibit the growth of *E. coli*).

Educational objective:

The minimal inhibitory concentration (MIC) gives the lowest concentration necessary to inhibit the growth of bacteria. A compound with the lowest MIC value in a set of tested compounds indicates that the compound has the greatest antimicrobial activity of the set.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401274

System:
Functional Groups and Their Reactions